Unit 15

Molarity, pH scale and strengths of acids and alkalis

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Molarity
- pH scale
- Strengths of acids and alkalis

15.1 Concentration of a solution

We can express the concentration of a solution in **molarity**.

✓ The molarity of a solution is the number of moles of solute dissolved in 1 dm^3 of the solution.

1 dm^3 = 1 000 cm^3

The units of molarity are mol dm^{-3} or M. We can calculate the molarity of a solution in the following way:

✓ Molarity of a solution (mol dm^{-3}) = $\frac{\text{number of moles of solute (mol)}}{\text{volume of solution } (dm^3)}$

Remember it is the amount of solute dissolved in 1 dm^3 of solution, NOT 1 dm^3 of water.

A **concentrated solution** is one in which the concentration of solute is high, say above 5 mol dm^{-3}.

A **dilute solution** is one in which the concentration of solute is low, say less than 2 mol dm^{-3}.

Fig. 15.1 shows the colour of copper(II) sulphate solutions of different concentrations.

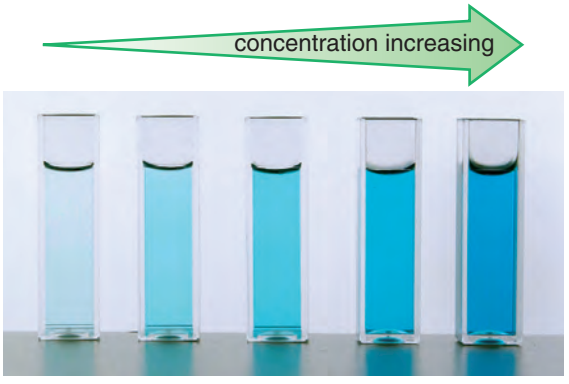


Fig. 15.1 Copper(II) sulphate solutions of different concentrations

molarity 摩爾濃度　　concentrated solution 濃溶液　　dilute solution 稀溶液

Example 15.1

Q 5.78 g of calcium chloride ($CaCl_2$) are dissolved in water and the solution is made up to 250.0 cm^3. Calculate the molarity of the solution.

(Relative atomic masses: Cl = 35.5, Ca = 40.1)

CE 2005 Paper 2 Q22
DSE 2012 Paper 1A Q10

A Molar mass of $CaCl_2$ = (40.1 + 2 x 35.5) g mol^{-1} = 111.1 g mol^{-1}

$$\text{Number of moles of } CaCl_2 = \frac{\text{mass}}{\text{molar mass}}$$
$$= \frac{5.78 \text{ g}}{111.1 \text{ g mol}^{-1}}$$
$$= 0.0520 \text{ mol}$$

$$\text{Volume of solution} = \frac{250.0}{1\,000} \text{ dm}^3$$

$$\text{Molarity of } CaCl_2 \text{ solution} = \frac{\text{number of moles of } CaCl_2}{\text{volume of solution}}$$
$$= \frac{0.0520 \text{ mol}}{\left(\frac{250.0}{1\,000}\right) \text{dm}^3}$$
$$= 0.208 \text{ mol dm}^{-3} \text{ (M)}$$

∴ the molarity of the calcium chloride solution is 0.208 mol dm^{-3} (M).

Example 15.2

Q What is the mass of sodium sulphate (Na_2SO_4) required to prepare 1.50 dm^3 of 1.60 mol dm^{-3} solution?

(Relative atomic masses: O = 16.0, Na = 23.0, S = 32.1)

A Molar mass of Na_2SO_4 = (2 x 23.0 + 32.1 + 4 x 16.0) g mol^{-1} = 142.1 g mol^{-1}

$$\text{Molarity of sodium sulphate solution} = \frac{\text{number of moles of } Na_2SO_4}{\text{volume of solution}}$$

$$1.60 \text{ mol dm}^{-3} = \frac{\text{number of moles of } Na_2SO_4}{1.50 \text{ dm}^3}$$

$$\text{Number of moles of } Na_2SO_4 = 1.60 \text{ mol dm}^{-3} \times 1.50 \text{ dm}^3$$
$$= 2.40 \text{ mol}$$

$$\text{Mass of } Na_2SO_4 \text{ required} = \text{number of moles of } Na_2SO_4 \times \text{molar mass of } Na_2SO_4$$
$$= 2.40 \text{ mol} \times 142.1 \text{ g mol}^{-1}$$
$$= 341 \text{ g}$$

∴ 341 g of sodium sulphate are required.

Example 15.3

Q 25.0 cm^3 of 0.250 mol dm^{-3} magnesium nitrate solution are mixed with 75.0 cm^3 of 0.100 mol dm^{-3} nitric acid. What is the concentration of nitrate ions in the resulting solution?

CE 2010 Paper 2 Q39
DSE 2014 Paper 1A Q6

A 1 mole of $Mg(NO_3)_2$ contains 2 moles of nitrate ions.

Number of moles of nitrate ions in magnesium nitrate solution

$= 2 \times 0.250 \text{ mol dm}^{-3} \times \frac{25.0}{1\,000} \text{ dm}^3$

$= 0.0125$ mol

1 mole of HNO_3 contains 1 mole of nitrate ions.

Number of moles of nitrate ions in nitric acid $= 0.100 \text{ mol dm}^{-3} \times \frac{75.0}{1\,000} \text{ dm}^3$

$= 0.00750$ mol

Total number of moles of nitrate ions $= (0.0125 + 0.00750)$ mol

$= 0.0200$ mol

Total volume of resulting solution $= (25.0 + 75.0)$ cm^3

$= 100.0$ cm^3

Concentration of nitrate ions in resulting solution $= \frac{0.0200 \text{ mol}}{\left(\frac{100.0}{1\,000}\right) \text{ dm}^3}$

$= 0.200 \text{ mol dm}^{-3}$

$\therefore$ the concentration of nitrate ions in the resulting solution is 0.200 mol dm^{-3}.

Practice 15.1

Refer to page T24.

1 26.0 g of ammonium nitrate (NH_4NO_3) were dissolved in 500.0 cm^3 of solution. Calculate the molarity of the solution.

0.650 mol dm^{-3} (M)

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0)

2 What is the mass of solute present in 2.50 dm^3 of 0.200 mol dm^{-3} potassium carbonate (K_2CO_3) solution?

69.1 g

(Relative atomic masses: C = 12.0, O = 16.0, K = 39.1)

Do you know

The origin of the term pH

The term pH is derived from the French term *puissance d'hydrogène* (or in English, the 'power of hydrogen') and refers to the power of 10 (the exponent) used to express the concentration of hydrogen ions.

15.2 The pH scale

The concentration of hydrogen ions in aqueous solutions commonly ranges from 2 mol dm^{-3} to 10^{-14} mol dm^{-3}. The concentration of hydrogen ions in dilute hydrochloric acid is about 100 000 000 000 000 (i.e. 10^{14}) times greater than the concentration of hydrogen ions in dilute sodium hydroxide solution.

Given such a wide range of concentrations, chemists find it convenient to use a logarithmic scale to measure the concentration of hydrogen ions in solutions. This is the **pH scale**.

✓ The pH of a solution is $-\log_{10}$ of the molar concentration of hydrogen ions in that solution.

✓ $pH = -\log_{10}[H^+]$

Square brackets [ ] are used to denote the molar concentration of a species.

Using the following formula, we can convert a pH value back to the concentration of hydrogen ions:

✓ $[H^+] = 10^{-pH}$

Because pH values are logarithms, they have no units.

The pH of a **neutral** solution is 7. A solution with a pH less than 7 is acidic while a solution with a pH greater than 7 is alkaline.

✓ Acidic solution $pH < 7$

✓ Neutral solution $pH = 7$

✓ Alkaline solution $pH > 7$

The pH of a neutral solution is 7 when measured at 298 K.

pH scale pH 標度　　neutral 中性的

Example 15.4

Q A 0.0500 mol dm^{-3} sulphuric acid sample was prepared. Assuming that the acid dissociated completely in water, calculate the pH of the acid sample.

A Sulphuric acid dissociates completely according to the following equation:

$$\underset{0.0500 \text{ mol dm}^{-3}}{H_2SO_4(aq)} \longrightarrow \underset{? \text{ mol dm}^{-3}}{2H^+(aq)} + SO_4^{2-}(aq)$$

According to the equation, 1 mole of H_2SO_4 dissociates to give 2 moles of hydrogen ions.

i.e. concentration of hydrogen ions = 2 x 0.0500 mol dm^{-3}
= 0.100 mol dm^{-3}

pH of acid = $-\log_{10}(0.100)$
= $-(-1.00)$ = 1.00

∴ the pH of the sulphuric acid sample is 1.00.

Example 15.5

Q The pH of human blood is 7.40. What is the concentration of hydrogen ions in blood?

A pH of blood = $-\log_{10}[H^+]$ = 7.40

i.e. $\log_{10}[H^+] = -7.40$
$[H^+] = 10^{-7.40}$
$= 3.98 \times 10^{-8}$ mol dm^{-3}

∴ the concentration of hydrogen ions in blood is 3.98×10^{-8} mol dm^{-3}.

Practice 15.2

1 Calculate the pH of a sample of milk whose concentration of hydrogen ions is 2.51×10^{-7} mol dm^{-3}.

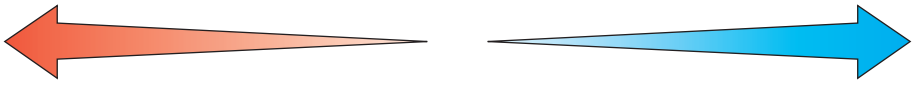
2 What is the concentration of hydrogen ions in a sample of acid rain with a pH of 3.44?

Practice 15.2

1 pH of milk = $-\log_{10}(2.51 \times 10^{-7})$
= 6.60

2 pH of acid rain = $-\log_{10}[H^+]$
= 3.44
i.e. $\log_{10}[H^+] = -3.44$
$[H^+] = 10^{-3.44}$
$= 3.63 \times 10^{-4}$ (mol dm^{-3})

The pH scale is used to measure the degree of acidity or **alkalinity** of a solution. The pH scale ranges from 0 to 14. Fig. 15.2 lists the concentrations of hydrogen ions in solutions with various pH values.



pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Concentration of hydrogen ions (mol dm^{-3})	10^0 (i.e. 1)	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}	10^{-9}	10^{-10}	10^{-11}	10^{-12}	10^{-13}	10^{-14}

Fig. 15.2 Concentrations of hydrogen ions in solutions with various pH values

Notice the following points from Fig. 15.2:

- as the concentration of hydrogen ions increases, the pH decreases;
- the lower the pH, the higher the acidity of the solution; and
- a change of one pH unit represents a tenfold change in the concentration of hydrogen ions.

Fig. 15.3 shows the pH values of some common acidic substances. The lower the pH, the more acidic is the substance. N1

N1 Refer to page T25.

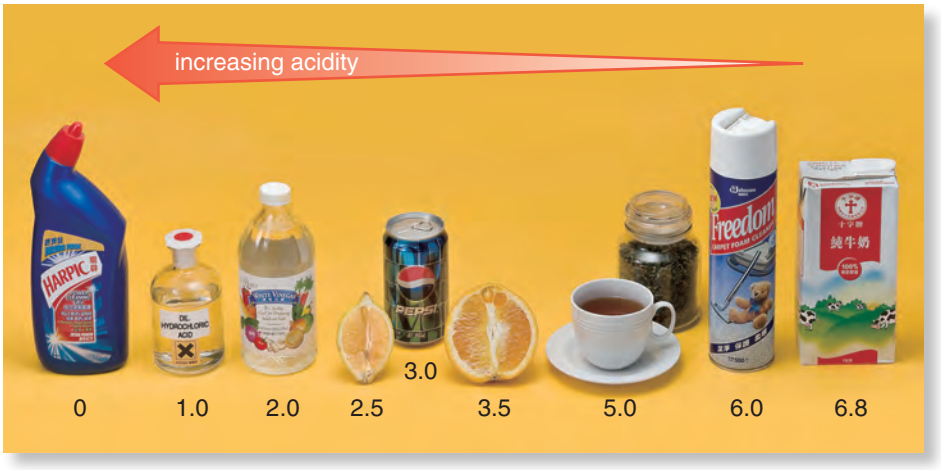


Fig. 15.3 The pH values of some common acidic substances

Do you know

pH of normal rainwater

As rain falls through the atmosphere, some carbon dioxide gas present in the atmosphere dissolves in the water. Carbonic acid ($H_2CO_3(aq)$) is formed as a result.

$$CO_2(g) \xrightarrow{H_2O(l)} CO_2(aq)$$

$$CO_2(aq) \xrightarrow{H_2O(l)} H_2CO_3(aq)$$

Hence normal rainwater is slightly acidic and has a pH of about 5.6.

alkalinity 鹼度

Fig. 15.4 shows the pH values of some common alkaline substances. The higher the pH, the more alkaline is the substance.



Fig. 15.4 The pH values of some common alkaline substances

N2 Some everyday uses of indicators:
- testing the acidity or alkalinity of soils (because some plants need an acidic soil, such as azaleas and camellias, while others need an alkaline soil, including most annual flowers and vegetables);
- testing home swimming pools (these need to be approximately neutral, though adding chemicals to sanitize the water can change its acid-alkali balance).

15.1

Classifying substances as acidic, alkaline or neutral using indicators.

15.3 Determining pH values of solutions

In the laboratory, we can use some simple methods to determine the pH values of different solutions. N2

Universal indicator solution



Fig. 15.5 A bottle of universal indicator solution

Universal indicator solution is a mixture of several indicators (Fig. 15.5). It changes colour gradually, from red through green to violet as the solution under test changes from acidic through neutral to alkaline.

To determine the pH of a solution, we add a few drops of universal indicator solution to the solution under test and compare the resulting colour to a standard pH colour chart such as the one shown in Fig. 15.6.

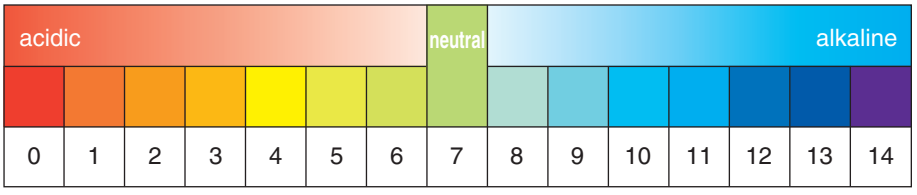


Fig. 15.6 pH colour chart

The colours of the pH colour chart vary according to the brand of indicator solution used.

universal indicator solution 通用指示劑溶液

We can also use *pH paper* (Fig. 15.7) to determine the pH of a solution.

pH papers are made by soaking strips of paper in a universal indicator solution followed by drying.



Fig. 15.7 A roll of pH paper

The pH meter

We can use a **pH meter** to determine the pH of a solution accurately (Fig. 15.8). Dip the electrode of the pH meter into a solution of unknown pH and read the pH directly from the digital display.

CE 2006 Paper 1 Q9(c)



Fig. 15.8 Using a pH meter to determine the pH of a solution

Do you know ?

Monitoring water quality

Measuring the pH of the fresh water supplied to households is an important part of the process for monitoring the quality of water supplied.

Ideally tap water should have a pH range of 6 to 8.

Reference Website

Refer to the following website for simulations of measuring the pH of solutions of acids, bases and salts:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/flashfiles/acidbasepH/ph_meter.html
(accessed July 2014)

Data-logger

We can also use a **data-logger** with a **pH sensor** to determine the pH of a solution accurately. Dip the pH sensor of the system into a solution of unknown pH and read the pH directly from the system display.

pH paper pH 試紙　pH meter pH 計　data-logger 數據收集儀　pH sensor pH 感應器

Animation
Strong and weak acids

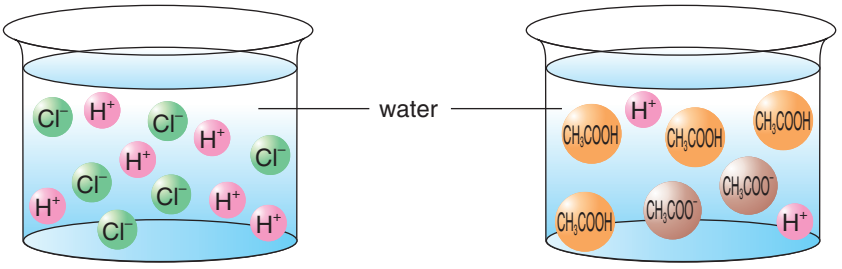
15.4 Strong and weak acids

When hydrogen chloride gas dissolves in water, almost all the hydrogen chloride molecules dissociate to give hydrogen ions and chloride ions. The hydrochloric acid formed contains mainly hydrogen ions ($H^+(aq)$), chloride ions ($Cl^-(aq)$) and water molecules (Fig. 15.9a). Hydrochloric acid is a **strong acid**.

$$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$$

When pure ethanoic acid dissolves in water, only a few molecules dissociate to give ions. There are four kinds of particles in ethanoic acid: hydrogen ions, ethanoate ions ($CH_3COO^-(aq)$), ethanoic acid molecules and water molecules (Fig. 15.9b). Ethanoic acid is a **weak acid**.

$$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$$



(a) (b)

Fig. 15.9 (a) Hydrochloric acid almost completely dissociates in water
(b) Ethanoic acid only partially dissociates in water
(This is only a schematic sketch. It does not show the actual numbers and sizes of particles in the solutions. The ions and molecules are colourless.)

✓ A strong acid is an acid that almost completely dissociates in water.

✓ A weak acid is an acid that only partially dissociates in water.

N3 Refer to page T25.

N3

ASL 2006 Q12(a)(i)

strong acid 強酸 weak acid 弱酸

Consider two acids HA and HB. HA is a strong acid. HB is a weak acid. We can use a graph format to show the concentrations of chemical species before and after dissociation in water for HA and HB respectively (Fig. 15.10).

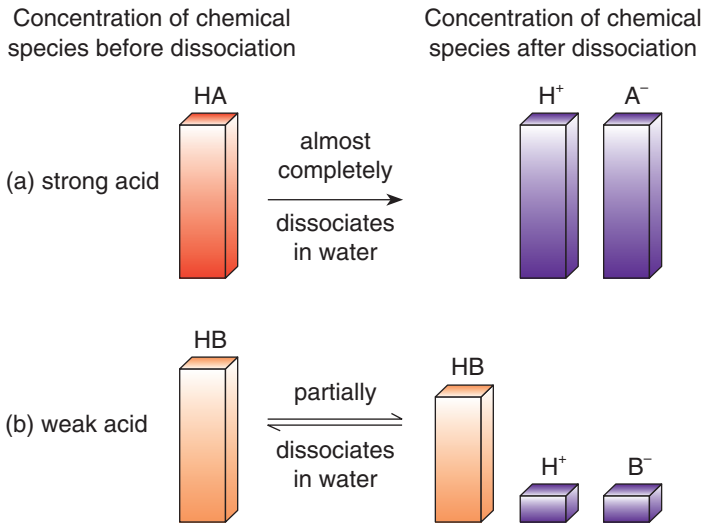


Fig. 15.10 (a) HA is a strong acid and it almost completely dissociates in water
(b) HB is a weak acid and it only partially dissociates in water

15.2
Distinguishing between a strong acid and a weak acid.

Water dissociates slightly to give hydrogen ions and hydroxide ions.

$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$

Hydrochloric acid, sulphuric acid and nitric acid are strong acids. Ethanoic acid and carbonic acid are weak acids. Table 15.1 shows the main types of particles present when some common acids dissolve in water.

Table 15.1

Main types of particles present when some common acids dissolve in water

	Acid	Chemical formula	Main types of particles present when the acid dissolves in water (besides water molecules and hydroxide ions)
Strong acids	hydrochloric acid	HCl	$H^+(aq)$, $Cl^-(aq)$
	sulphuric acid	H_2SO_4	$H^+(aq)$, $HSO_4^-(aq)$, $SO_4^{2-}(aq)$
	nitric acid	HNO_3	$H^+(aq)$, $NO_3^-(aq)$
Weak acids	ethanoic acid	CH_3COOH	$H^+(aq)$, $CH_3COO^-(aq)$, $CH_3COOH(aq)$
	carbonic acid	H_2CO_3	$H^+(aq)$, $HCO_3^-(aq)$, $CO_3^{2-}(aq)$, $CO_2(aq)$, $H_2CO_3(aq)$

15.5 Comparing the strengths of acids N4

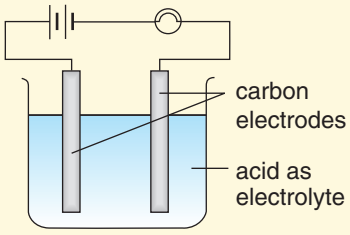
N4 Refer to page T26.

Video
Comparing the pH, electrical conductivity and reaction with magnesium of acids with different strength

Let us compare the pH, electrical conductivity and reaction with magnesium of two acids: 0.1 mol dm^{-3} hydrochloric acid and 0.1 mol dm^{-3} ethanoic acid. Table 15.2 shows the results of the tests with the deductions.

Table 15.2

Comparing the pH, electrical conductivity and reaction with magnesium of 0.1 mol dm^{-3} hydrochloric acid and 0.1 mol dm^{-3} ethanoic acid

Test	Results	Deduction
Measuring the pH DSE 2012 Paper 1A Q4	The pH of the hydrochloric acid is lower than that of the ethanoic acid.	The hydrochloric acid has a higher concentration of hydrogen ions than the ethanoic acid does.
Using each acid as an electrolyte in the set-up shown below:  	The bulb glows more brightly for the hydrochloric acid than for the ethanoic acid.	The electrical conductivity of a solution is proportional to the concentration of mobile ions. The results show that the hydrochloric acid has a higher concentration of mobile ions than the ethanoic acid does.
Adding identical magnesium ribbons to each acid.	The rate of evolution of gas bubbles for the hydrochloric acid is higher than that for the ethanoic acid.	In the reaction between magnesium and an acid, magnesium reacts with the hydrogen ions in the acid. The rate of the reaction increases as the concentration of hydrogen ions in the acid increases. The results show that the hydrochloric acid has a higher concentration of hydrogen ions than the ethanoic acid does.

N5 Examination questions often ask about methanoic acid and ethanoic acid.
- They are covalent compounds.
- They are electrolytes.
- They are both monobasic.

Hydrochloric acid is a strong acid. It almost completely dissociates in water to give hydrogen ions and chloride ions. Ethanoic acid is a weak acid. It only partially dissociates in water, forming very few hydrogen ions. N5

DSE 2012 Paper 1A Q4

CE 2009 Paper 2 Q36

Example 15.6

Q Consider the following information on two reactions involving magnesium ribbons of the same shape.

	Reacting mixture
Reaction 1	1.5 g of Mg + 100 cm^3 of 1.0 mol dm^{-3} HCl(aq)
Reaction 2	1.5 g of Mg + 100 cm^3 of 1.0 mol dm^{-3} H_2SO_4(aq)

The initial rate is the rate at the start of the reaction.

Would you expect the initial rate of *Reaction 2* to be higher, lower or the same as that of *Reaction 1*? Explain your answer.

A The initial rate of *Reaction 2* is higher than that of *Reaction 1*.

In the reaction between magnesium and the acids, magnesium reacts with the hydrogen ions in the acids.

Sulphuric acid is a dibasic acid while hydrochloric acid is a monobasic acid.

Both are strong acids and dissociate completely in water.

1.0 mol dm^{-3} H_2SO_4(aq) has a higher concentration of hydrogen ions than 1.0 mol dm^{-3} HCl(aq).

Thus, the initial rate of *Reaction 2* is higher than that of *Reaction 1*.

Worksheet

A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Explain whether you agree with the following statement.

'A is a stronger acid than B, so the pH of an aqueous solution of A must be lower than that of B.'

Discussion

No.

'A is a stronger acid than B' only means that the degree of dissociation of A is larger than that of B.

However, the pH of an aqueous solution of an acid depends on both its degree of dissociation and concentration.

Thus, an aqueous solution of the stronger acid A may have a higher pH than that of the weaker acid B if the concentration of acid B is higher than that of acid A by an adequate amount.

15.6 Strong and weak alkalis

Animation
Strong and weak alkalis

Alkalis, like acids, can also be classified as strong and weak ones.

✓ A **strong alkali** is an alkali that almost completely dissociates in water to give hydroxide ions ($OH^-(aq)$).

✓ A **weak alkali** is an alkali that only partially dissociates in water to give very few hydroxide ions ($OH^-(aq)$).

Sodium hydroxide dissociates almost completely into sodium ions and hydroxide ions when it dissolves in water (Fig. 15.11a). The solution contains mainly sodium ions, hydroxide ions and water molecules.

$$NaOH(aq) \longrightarrow Na^+(aq) + OH^-(aq)$$

When ammonia gas dissolves in water, it reacts with water to give ammonium ions and hydroxide ions. However, ammonia does not react with water completely. Only a small amount of hydroxide ions are formed (Fig. 15.11b).

$$NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

Thus, sodium hydroxide is a strong alkali while ammonia is a weak alkali.

All hydroxides of Group I metals are strong alkalis.

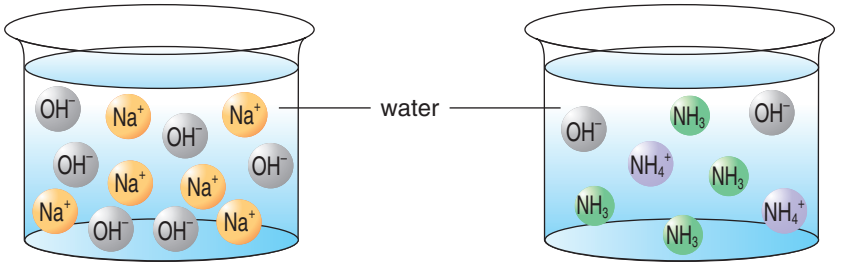


Fig. 15.11 (a) Sodium hydroxide almost completely dissociates in water
(b) Ammonia does not react with water completely
(This is only a schematic sketch. It does not show the actual numbers and sizes of particles in the solutions. The ions and molecules are colourless.)

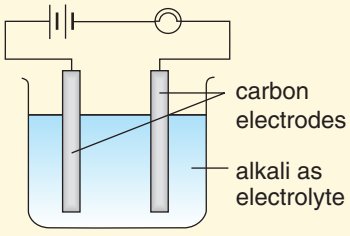
strong alkali 強鹼　　weak alkali 弱鹼

15.7 Comparing the strengths of alkalis N6

Let us compare the pH and electrical conductivity of two alkalis: 0.1 mol dm^{-3} sodium hydroxide solution and 0.1 mol dm^{-3} aqueous ammonia. Table 15.3 shows the results of the tests with the deductions.

Table 15.3

Comparing the pH and electrical conductivity of 0.1 mol dm^{-3} sodium hydroxide solution and 0.1 mol dm^{-3} aqueous ammonia

Test	Results	Deduction
Measuring the pH	The pH of the sodium hydroxide solution is higher than that of the aqueous ammonia.	The sodium hydroxide solution is more alkaline than the aqueous ammonia.
Using each alkali as an electrolyte in the set-up shown below: 	The bulb glows more brightly for the sodium hydroxide solution than for the aqueous ammonia.	The sodium hydroxide solution has a higher concentration of mobile ions than the aqueous ammonia does.

Sodium hydroxide is a strong alkali. It almost completely dissociates in water to give sodium ions and hydroxide ions. Ammonia is a weak alkali. It forms very few hydroxide ions when dissolved in water.

CE 2009 Paper 2 Q37

N6 Comparing strong alkalis and weak alkalis

Examination questions may ask students to compare strong alkalis with weak alkalis:

e.g. comparing 0.5 mol dm^{-3} NaOH(aq) and 0.5 mol dm^{-3} NH_3(aq)

- they contain different number of hydroxide ions;
- they have different pH values;
- they have different electrical conductivity;
- 0.5 mol dm^{-3} NaOH(aq) gives a larger temperature rise when completely neutralized by 1 mol dm^{-3} HCl(aq);
- both react with $FeSO_4$(aq), each giving a green precipitate.

15.8 Strength versus concentration

CE 2008 Paper 2 Q30
DSE 2014 Paper 1B Q5(b)

When we talk about the strength of an acid, we are referring to the extent of its dissociation in water. When we talk about the concentration of an acid, we are referring to the amount of the acid in a unit volume of solution.

Consider a 5 mol dm^{-3} hydrochloric acid and a 5 mol dm^{-3} ethanoic acid. Both of them are concentrated acids. However, hydrochloric acid is a strong acid while ethanoic acid is a weak acid. Hence 5 mol dm^{-3} hydrochloric acid is a concentrated solution of a strong acid while 5 mol dm^{-3} ethanoic acid is a concentrated solution of a weak acid (Fig. 15.12).

Similarly, 0.1 mol dm^{-3} hydrochloric acid is a dilute solution of a strong acid while 0.1 mol dm^{-3} ethanoic acid is a dilute solution of a weak acid (Fig. 15.12).

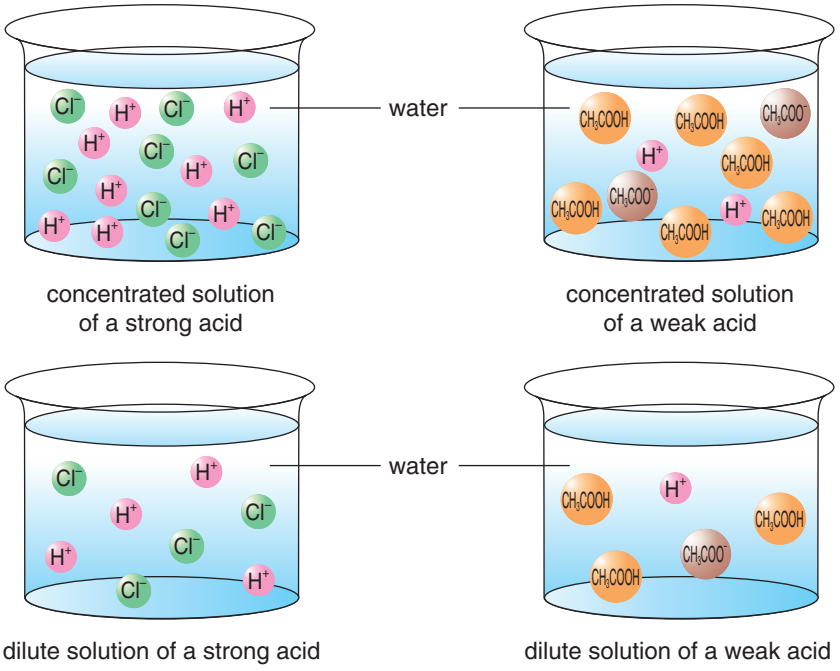


Fig. 15.12 Comparing the strengths and concentrations of acids (This is only a schematic sketch. It does not show the actual numbers and sizes of particles in the solutions. The ions and molecules are colourless.)

Practice 15.3

1 mol dm^{-3} sulphuric acid < 1 mol dm^{-3} hydrochloric acid < 0.1 mol dm^{-3} nitric acid < 0.1 mol dm^{-3} ethanoic acid

Both sulphuric acid and hydrochloric acid are strong acids and dissociate completely.

$H_2SO_4(aq) \longrightarrow 2H^+(aq) + SO_4^{2-}(aq)$

$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$

Every HCl molecule dissociates to give one hydrogen ion while every H_2SO_4 molecule dissociates to give two hydrogen ions.

Thus, the concentration of hydrogen ions in 1 mol dm^{-3} sulphuric acid is higher than that in 1 mol dm^{-3} hydrochloric acid. Hence the pH of 1 mol dm^{-3} sulphuric acid is lower than that of 1 mol dm^{-3} hydrochloric acid.

The concentration of hydrogen ions in 1 mol dm^{-3} hydrochloric acid is higher than that in 0.1 mol dm^{-3} nitric acid. Hence the pH of 1 mol dm^{-3} hydrochloric acid is less than that of 0.1 mol dm^{-3} nitric acid.

Ethanoic acid is a weak acid. It only partially dissociates in water.

$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$

The concentration of hydrogen ions in 0.1 mol dm^{-3} ethanoic acid is lower than that in 0.1 mol dm^{-3} nitric acid, a strong acid. Hence the pH of 0.1 mol dm^{-3} ethanoic acid is higher than that of 0.1 mol dm^{-3} nitric acid.

Practice 15.3

DSE 2013 Paper 1A Q10

Arrange the acids below in the order of increasing pH and explain your answer.

0.1 mol dm^{-3} ethanoic acid, 0.1 mol dm^{-3} nitric acid, 1 mol dm^{-3} sulphuric acid, 1 mol dm^{-3} hydrochloric acid

Have you mastered?

Key terms

molarity 摩爾濃度 40 concentrated solution 濃溶液 40 dilute solution 稀溶液 40

pH scale pH 標度 43 neutral solution 中性溶液 43 acidic solution 酸性溶液 43

alkaline solution 鹼性溶液 43 alkalinity 鹼度 45 universal indicator solution 通用指示劑溶液 46

pH meter pH 計 47 data-logger 數據收集儀 47 pH sensor pH 感應器 47

strong / weak acid 強 / 弱酸 48 strong / weak alkali 強 / 弱鹼 52

Checklist

After studying this unit, you should be able to

- ☐ relate the pH scale to the acidity or alkalinity of substances;
- ☐ perform calculations related to the concentration of hydrogen ions and the pH of a solution of a strong acid;
- ☐ suggest appropriate ways to determine the pH of substances;
- ☐ relate the strength of acids and alkalis to their extent of dissociation;
- ☐ suggest experiments to compare the strength of acids or alkalis.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

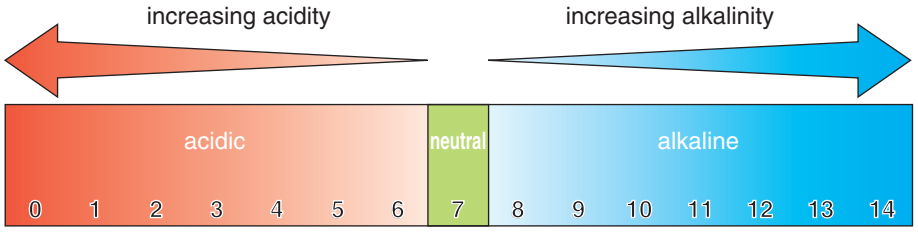
1 The molarity of a solution is the number of moles of solute dissolved in 1 dm^3 of the solution.

$$\text{Molarity of a solution (mol } dm^{-3}) = \frac{\text{number of moles of solute (mol)}}{\text{volume of solution } (dm^3)}$$

2 The pH of a solution is $-\log_{10}$ of the molar concentration of hydrogen ions in that solution.

$$pH = -\log_{10}[H^+]$$

3 The pH scale is used to measure the degree of acidity or alkalinity of a solution. Its value ranges from 0 to 14.



4 Methods used to determine the pH values of solutions include:

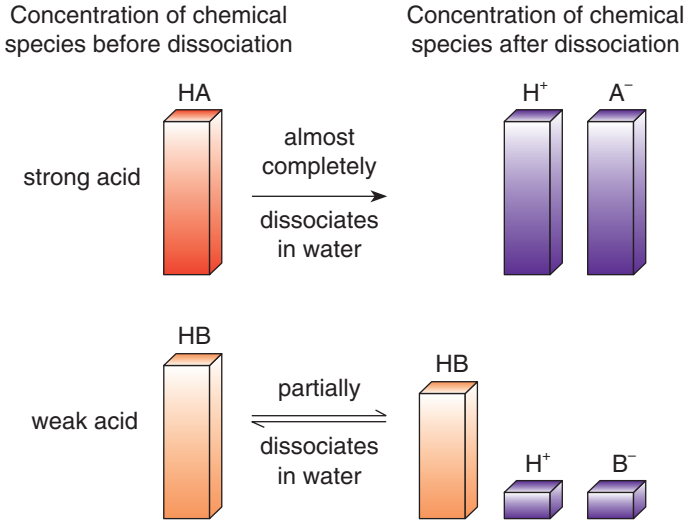
a) using universal indicator solution;

b) using a pH meter; and

c) using a data-logger with a pH sensor.

5 a) A strong acid is an acid that almost completely dissociates in water.

A weak acid is an acid that only partially dissociates in water.



b) Compared to a weak acid of the same concentration, a strong acid has

- a lower pH;
- a higher electrical conductivity;
- a higher rate of reaction with metals.

Have you mastered?

6 a) A strong alkali is an alkali that almost completely dissociates to give hydroxide ions ($OH^-(aq)$) in water.

A weak alkali is an alkali that only partially dissociates to give hydroxide ions in water.

b) Compared to a weak alkali of the same concentration, a strong alkali has

- a higher pH;
- a higher electrical conductivity.

7 The strength of an acid or alkali concerns the extent of dissociation of the acid or alkali in water.

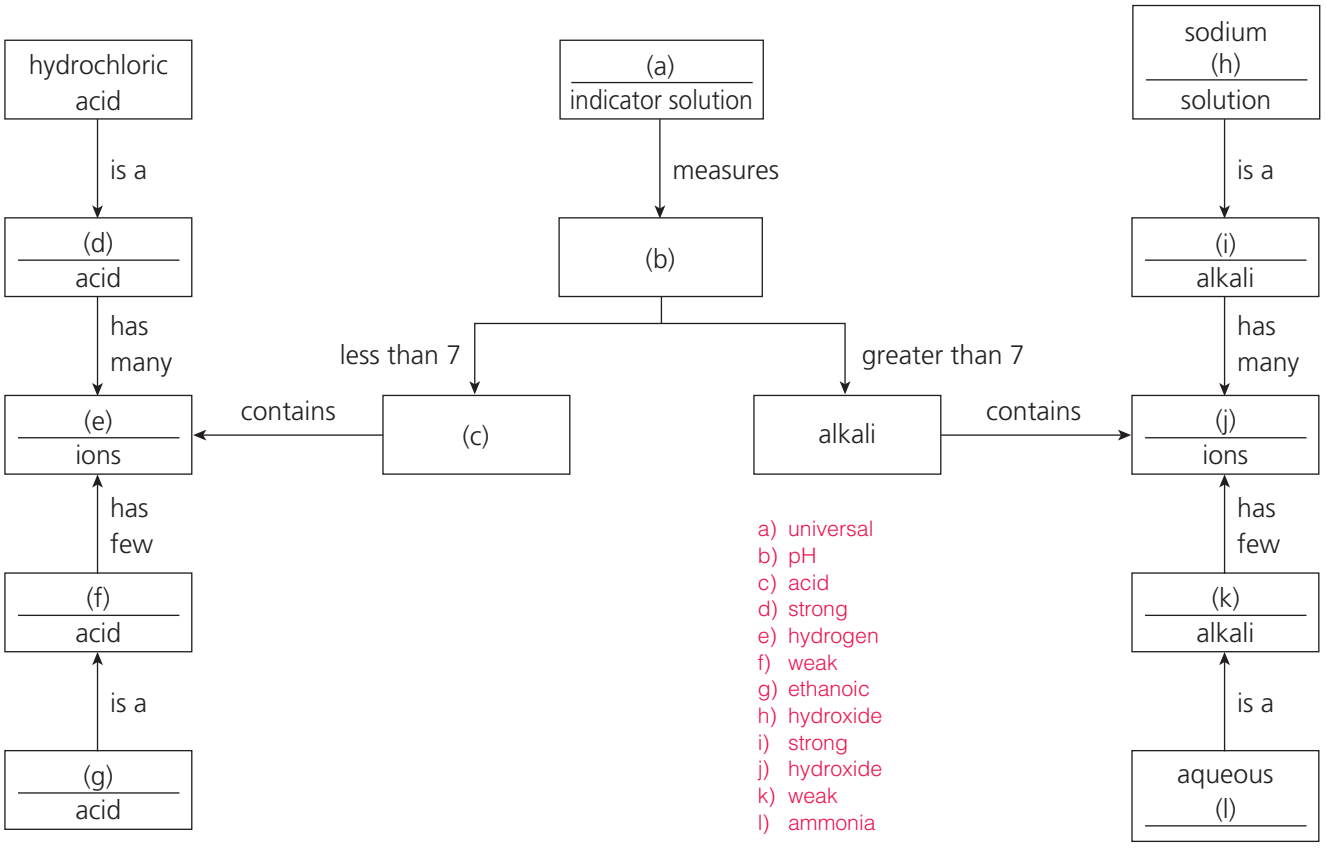
The concentration of an acid or alkali concerns the amount of the acid or alkali in a unit volume of solution.

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) universal
b) pH
c) acid
d) strong
e) hydrogen
f) weak
g) ethanoic
h) hydroxide
i) strong
j) hydroxide
k) weak
l) ammonia

2 a) Molarity of a solution

$$= \frac{\text{number of moles of solute (mol)}}{\text{volume of solution (dm}^3\text{)}}$$

b) The pH scale is used to measure the degree of acidity or alkalinity of a solution.

c) Solutions with a pH less than 7 are acidic.

d) Solutions with a pH greater than 7 are alkaline.

e) Solutions with a pH of 7 are neutral.

f) An acid that almost completely dissociates in water is a strong acid.

g) An acid that only partially dissociates in water is a weak acid.

3 Five substances A to E are tested with moist pH paper. The results are shown below.

Substance	A	B	C	D	E
pH	3	1	7	13	9

Which substance is

a) neutral?
C
b) strongly acidic?
B
c) weakly acidic?
A
d) strongly alkaline?
D
e) weakly alkaline?
E

Part II Multiple choice questions

4 A sample of 1.02 g of potassium hydrogenphthalate ($C_8H_5O_4K$) is dissolved completely in distilled water, and then diluted to 250.0 cm^3. What is the concentration of the solution obtained?

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, K = 39.1)

A 0.004 M
B 0.010 M
(C) 0.020 M
D 4.080 M

(HKDSE, Paper 1A, 2012, 10)

5 What is the mass of ethanedioic acid crystals ($(COOH)_2 \bullet 2H_2O$) required to prepare 500.0 cm^3 of 0.120 mol dm^{-3} solution?

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

(A) 7.56 g
B 15.1 g
C 47.6 g
D 52.5 g

$$\text{Molarity of the acid solution} = \frac{\text{number of moles of } (COOH)_2 \bullet 2H_2O}{\text{volume of solution}}$$

$$0.120 \text{ mol } dm^{-3} = \frac{\text{number of moles of } (COOH)_2 \bullet 2H_2O}{\left(\frac{500.0}{1\,000}\right) dm^3}$$

$$\text{Number of moles of } (COOH)_2 \bullet 2H_2O = 0.120 \text{ mol } dm^{-3} \times \frac{500.0}{1\,000} dm^3$$

$$= 0.0600 \text{ mol}$$

$$\text{Mass of } (COOH)_2 \bullet 2H_2O \text{ required} = 0.0600 \text{ mol} \times 126.0 \text{ g mol}^{-1}$$

$$= 7.56 \text{ g}$$

∴ 7.56 g of ethanedioic acid crystals are required.

6 20.0 cm^3 of 1.0 M NaCl(aq) are mixed with 10.0 cm^3 of 2.0 M Na_2CO_3(aq). What is the concentration of Na^+(aq) ions in the resulting solution?

A 1.3 M
B 1.5 M
(C) 2.0 M
D 3.0 M

(HKCEE, Paper 2, 2010, 39)

7 Which of the following combinations is correct?

	Substance	pH
(A)	car battery acid	1
B	gastric juice	10
C	limewater	14
D	soda water	11

Car battery acid is sulphuric acid.

8 At 298 K, the pH of 0.10 mol dm^{-3} HCl(aq) is 1. Which of the following statements is correct?

A At 298 K, the pH of 0.20 mol dm^{-3} HCl(aq) is 2.
B At 298 K, the pH of 0.20 mol dm^{-3} HCl(aq) is 0.5.
(C) At 298 K, the pH of 0.01 mol dm^{-3} HCl(aq) is 2.
D At 298 K, the pH of 0.01 mol dm^{-3} HCl(aq) is 0.1.

(HKDSE, Practice paper 1A, 2012, 8)

9 Consider the four solutions W, X, Y and Z listed below:

W: 0.01 mol dm^{-3} HNO_3(aq)
X: 0.01 mol dm^{-3} H_2SO_4(aq)
Y: 0.01 mol dm^{-3} KOH(aq)
Z: 0.10 mol dm^{-3} KOH(aq)

Which of the following represents the four solutions arranged in increasing order of pH?

A W, X, Y, Z
B W, X, Z, Y
(C) X, W, Y, Z
D X, W, Z, Y

(HKDSE, Paper 1A, 2013, 10)

10 Comparing the same volume of 0.5 M NaOH(aq) and 0.5 M NH_3(aq), which of the following is NOT correct?

	0.5 M NaOH(aq)	0.5 M NH_3(aq)
A	higher pH	lower pH
B	higher electrical conductivity	lower electrical conductivity
(C)	forms precipitate with $FeSO_4$ solution	does not form precipitate with $FeSO_4$ solution
D	larger temperature rise when completely neutralized by 1 M HCl	smaller temperature rise when completely neutralized by 1 M HCl

(HKCEE, Paper 2, 2009, 37)

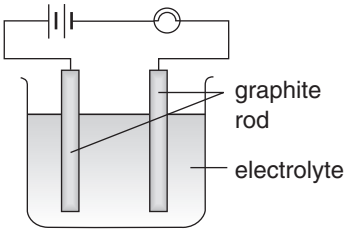
11 Which of the following statements concerning citric acid are correct?

(1) It is a weak acid.
(2) It is present in grapefruits.
(3) It is an electrolyte.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

12 Which of the following methods can be used to distinguish between 0.1 mol dm^{-3} HCl(aq) and 0.10 mol dm^{-3} CH_3CO_2H(aq)?

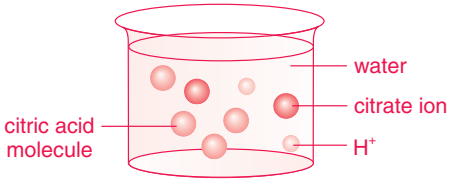
(1) Add magnesium ribbon of the same length to each solution and compare the rate of evolution of gas bubbles.
(2) Add 10 cm^3 of 0.1 mol dm^{-3} NaOH(aq) to 10 cm^3 of each solution and compare the temperature change.
(3) Use each solution as electrolyte in the set-up shown below and compare the brightness of the bulb.



A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

(HKDSE, Practice paper 1A, 2012, 24)

11 (1) Citric acid is a weak acid. When citric acid crystals dissolve in water, only a few citric acid molecules dissociate to give hydrogen ions.



(3) Citric acid exists as a solid at room temperature. Its aqueous solution can conduct electricity.

Part III Structured questions

13 Calculate the molarity of the following solutions:
Refer to page T27.
a) 10.1 g of KOH in 250.0 cm^3 solution
0.720 mol dm^{-3}
b) 7.93 g of $(NH_4)_2SO_4$ in 750.0 cm^3 solution
0.0800 mol dm^{-3}
c) 57.1 g of $FeCl_2$ in 2.50 dm^3 solution
0.180 mol dm^{-3}
(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, S = 32.1, Cl = 35.5, K = 39.1, Fe = 55.8)

14 What is the mass of solute present in 200.0 cm^3 of 1.20 mol dm^{-3} copper(II) sulphate solution?
38.3 g; refer to page T28.
(Relative atomic masses: O = 16.0, S = 32.1, Cu = 63.5)

15 20.0 cm^3 of 0.400 mol dm^{-3} ammonium carbonate solution are mixed with 40.0 cm^3 of 0.500 mol dm^{-3} ammonium nitrate solution.
0.600 mol dm^{-3}; refer to page T28.
What is the concentration of ammonium ions in the resulting solution?

16 Calculate the pH of
Refer to page T28.
a) 1.00×10^{-4} mol dm^{-3} hydrochloric acid;
4.00
b) 0.0480 mol dm^{-3} sulphuric acid.
1.02

17 What is the concentration of hydrogen ions in each of the following solutions?
Refer to page T29.
a) Coffee with a pH of 5.40
3.98×10^{-6} mol dm^{-3}
b) Saliva with a pH of 6.70
2.00×10^{-7} mol dm^{-3}

18 A 10.0 cm^3 sample of domestic sewage has an effluent pH of 6.80. After this sewage has been treated, the effluent pH is 7.00. What is the change in the concentration of hydrogen ions?

18 pH of effluent before treatment = 6.80
pH of effluent after treatment = 7.00
Concentration of hydrogen ions in effluent before treatment
$= 10^{-6.80}$
$= 1.58 \times 10^{-7}$ mol dm^{-3}
Concentration of hydrogen ions in effluent after treatment
$= 10^{-7.00}$ mol dm^{-3}
Change in concentration of hydrogen ions
$= (10^{-7} - 1.58 \times 10^{-7})$ mol dm^{-3}
$= -0.58 \times 10^{-7}$ mol dm^{-3}
$\therefore$ the concentration of hydrogen ions in the effluent decreases by 0.58×10^{-7} mol dm^{-3} upon treatment.

19 Two pieces of magnesium ribbon of equal mass were added to 50 cm^3 of 1 mol dm^{-3} hydrochloric acid and 50 cm^3 of 1 mol dm^{-3} ethanoic acid separately. The acid was in excess in each case. The information is shown in the table below.
Refer to page T29.

Reaction	Reactants	Time required for reaction to complete (s)
I	magnesium + 50 cm^3 of 1 mol dm^{-3} hydrochloric acid	200
II	magnesium + 50 cm^3 of 1 mol dm^{-3} ethanoic acid	?

a) How could you know when the reactions were complete?

b) Write an equation for the reaction between magnesium and dilute hydrochloric acid.

c) Would you expect the time required for the completion of *Reaction II* to be shorter or longer than that for *Reaction I*? Explain your answer.

20 Hydrogen fluoride and hydrogen chloride dissolve in water to give hydrofluoric acid and hydrochloric acid. Hydrofluoric acid is a weak acid but hydrochloric acid is a strong acid.
Refer to page T29.
a) What is meant by the term 'weak acid'?

b) Write the equation for the reaction between hydrofluoric acid and solid sodium carbonate, Na_2CO_3.

c) Write the ionic equation for the reaction between hydrofluoric acid and solid sodium carbonate, Na_2CO_3.

d) Describe how the reaction with Na_2CO_3 can be used to show that hydrochloric acid is strong and hydrofluoric acid is weak.

(OCR Advanced Subsidiary GCE, Chemistry, 2813/01, Jan. 2009, 2(d))

21 Hydrochloric acid, HCl, is a strong acid. Ethanoic acid, CH_3COOH, is a weak acid.

a) Hydrochloric acid completely dissociates in water.

$HCl \longrightarrow H^+ + Cl^-$

Ethanoic acid, CH_3COOH, only partially dissociates in water to make a mixture.

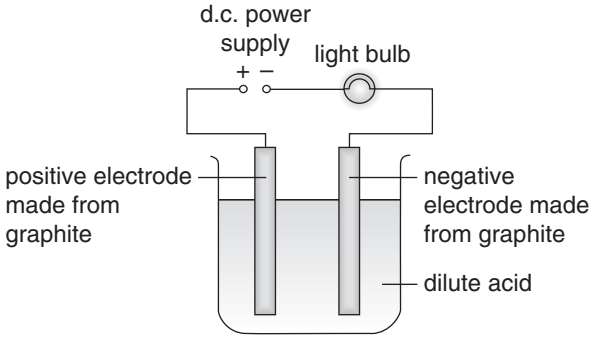
Write an equation to show this dissociation of ethanoic acid.

b) Ellen investigates the properties of these two acids. In each experiment Ellen uses the same concentration of acid. Look at Ellen's results table.

Test	Result with dilute hydrochloric acid	Result with dilute ethanoic acid
pH value	1	4
Reaction with magnesium ribbon	bubbles rapidly to make hydrogen	bubbles slowly to make hydrogen

Dilute hydrochloric acid reacts much faster than dilute ethanoic acid. Explain why.

c) Ellen also investigates the electrical conductivity of the two acids. Look at the apparatus she uses.



Ellen tests dilute ethanoic acid. She then tests dilute hydrochloric acid. Ellen uses the same concentration of both acids. She finds that the light bulb glows much more brightly with dilute hydrochloric acid. Explain why. Use ideas about ions.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2010, 6)

22 Go Grease is a drain and oven cleanser.



The active ingredient in Go Grease is the alkali sodium hydroxide (NaOH).

a) Name or give the formula of the ion that makes solutions alkaline.

b) Sodium hydroxide is a strong alkali.

In terms of dissociation, what is meant by the word 'strong'?

c) You are given solutions of sodium hydroxide and ammonia of the same concentration.

Describe and give the results of a test to show that sodium hydroxide is a stronger alkali than ammonia solution.

(AQA GCSE (Higher Tier), Chemistry, Unit C3, Jan. 2010, 3)

21 a) $CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$
b) Dilute hydrochloric acid has a higher concentration of hydrogen ions than dilute ethanoic acid does.
c) Dilute hydrochloric acid has a higher concentration of mobile ions.

22 a) Hydroxide ion / $OH^-(aq)$
b) Almost completely dissociate
c) Any one of the following:
- Measure the pH of each solution.
  The sodium hydroxide solution has a higher pH.
- Measure the electrical conductivity of each solution.
  The sodium hydroxide solution conducts better.